



# FlipFit

Where Fitness Meets Fun

**Group-OMEGA**



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# TEAM MEMBERS (OMEGA)

- Shrashti Jain (Leader)
- Ishita Sharma (Sub-Leader)
- Mansa Mahendru
- Devansh Narang
- Souvik Sarkar
- Shreya Dokania
- Rishi Nitin Godre
- Garapati Saranya
- Krishna Vilas Narwade
- Krishna Khandelwal

# PROBLEM STATEMENT

Design an intuitive and robust backend system for a Gym Management System for Flipkart's enterprise app,  
**FLIPFIT.**

# Framework for 1 week

01 Daily task planning

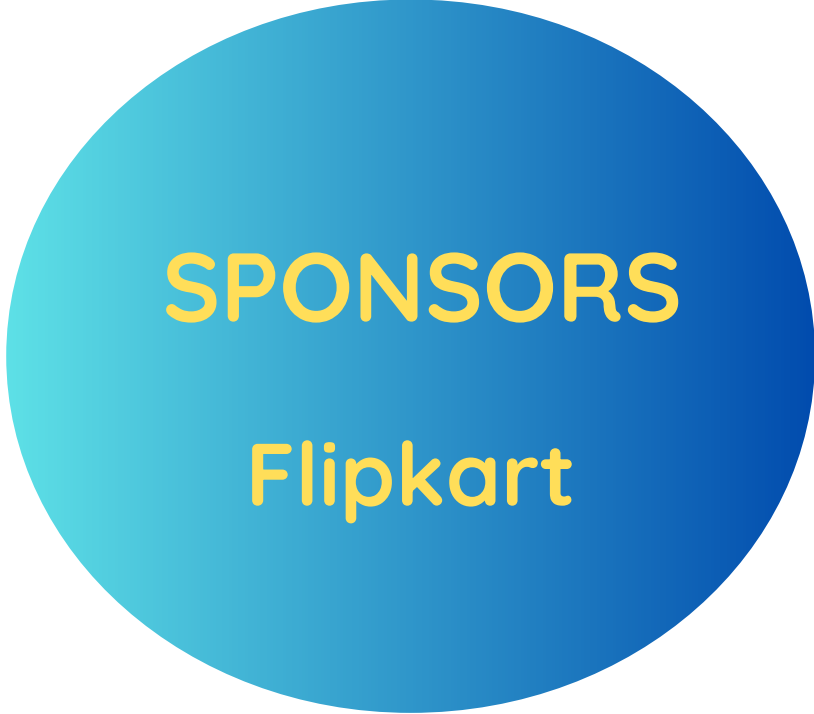
02 Learning new topic  
and technologies

03 Issue resolution and  
debugging

04 Project implementation as  
per UML



# STAKEHOLDERS



**SPONSORS**

**Flipkart**



**SME**

**Amit Balyan**

# ROADMAP



# PROJECT GOALS

Build a scalable backend for gym operations.

Manage gym centers and time slots efficiently

Enable smooth slot booking and cancellation

Prevent overbooking with proper seat control

Store data securely and reliably

Use a modular and well-structured architecture



# ENGINEERING PRACTICES

**1**

Modular design

**2**

Code refactoring

**3**

Exception handling

**4**

Code documentation

**5**

Git version control

**6**

Clean coding and  
Performance tuning

# TECH STACK



Java



Spring Tool Suite



Git



Dropwizard



Github



Postman

# UML Diagrams



[Github Repo](#)

# SCHEMA

## user

- userId
- name
- email
- phone

## gymCustomer

- customerId
- name
- address
- email
- phone
- password

## Admin

- adminId
- name
- email
- phone

## gymOwner

- ownerId
- name
- email
- phone
- pan
- address
- isApproved
- password

## booking

- bookingId
- bookingDate
- status

## bookingDetails

- bookingId
- userId
- centerId
- slotId
- bookingTime

## gymCenter

- centerId
- name
- isApproved
- location
- city

## payment

- paymentId
- amount
- mode
- status

# SCHEMA

## waitlist

- centerId
- waitlistId
- slotId

## Scheduler

- schedulerId
- date
- startTime
- endTime

## notification

- notificationId
- message
- type

## slot

- slotId
- capacity
- bookedSeats



# HOW WE PROCEEDED?

1

Analyzed requirements and prepared UML diagrams

2

Designed system structure based on project flow

3

Developed core Java backend modules

4

Applied Stream API and custom exception handling

5

Integrated MySQL database for data storage

6

Tested functionalities and fixed issues

# CHALLENGES AND LEARNINGS

01

Booking conflicts and data consistency

02

Integrating backend modules with MySQL database smoothly

03

Debugging complex workflows and handling edge cases

04

Designing a modular and scalable backend architecture

05

Applying real-world business logic using Java and Stream API

06

Meeting tight deadlines

DEMO



ANY  
QUESTIONS?

Thank You